Hooke’s law

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Connect spring displacement with restoring force.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 11.*
3 J of rotational kinetic energy. A rolling wheel also spins.

02. Worked example: explain the model
Displacement away from equilibrium; force points back, with 3 N magnitude.

03. Read and interpret the model
Stretched right gives force left; compressed left gives force right.

04. Apply the physics
3 N and 6 N.

05. Explain your reasoning
Force doubles; k is constant within linear range.

06. Review: connect an earlier idea *Review: Unit 4, Day 2.*
+35 J; force and displacement point the same way.

07. Exit check
8 N back toward natural length.

08. STRETCH: test the model
0.08 and 0.04 m; higher k needs less stretch for the same force.

Hooke’s law

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Connect spring displacement with restoring force.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 11.*
$v=\sqrt{(2\ gh)}\approx 3.464$ m/s.

02. Worked example: explain the model
Displacement away from equilibrium; force points back, with 3 N magnitude.

03. Read and interpret the model
Stretched right gives force left; compressed left gives force right.

04. Apply the physics
k 196 N/m; force 23.52 N.

05. Explain your reasoning
Force doubles; k is constant within linear range.

06. Review: connect an earlier idea *Review: Unit 4, Day 2.*
F=6 N; zero displacement gives 0 J.

07. Exit check
+12 N, right.

08. STRETCH: test the model
k 150 N/m; F=+6 N. Sign encodes restoring direction.